Natural Language Processing Project Guidelines

Academic year 2026–2027 First semester

Course code and section _____ Instructor _____

Assessment Specifications

The project should demonstrate a rigorous and well-justified NLP investigation, with sound implementation, evaluation, and critical analysis of the results.

Form a team of three students and choose one of the four project options in this document. Develop a working NLP notebook, investigate a focused question, and present your implementation and results in a project report and presentation.

The project is worth 25 course marks. Each student receives the group score out of 20 plus an individual score out of 5 for presentation, questions and teamwork. Use the requirements and assessment table in this document for every option.

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What your project should achieve

Your system must accept unfamiliar text and return a useful prediction. Your report should explain why the task matters, how you prepared the data, how two methods compare, and what the results show. A thoughtful explanation of limitations is part of a strong submission.

The initial proposal and first draft are due in Week 7, final files in Week 14, and presentations in Week 15. Exact deadlines and presentation slots will be posted on Blackboard.

Project Options

Choose one option. All four use the same report outline, submission requirements and grading criteria. The possible focuses below are examples to guide your proposal.

Option 1 Spam Detection

Build a notebook that accepts a message and returns Spam or Not spam. For example, “You won a cash prize. Pay a fee now to claim it” should be treated as spam, while “Please bring your notebook to tomorrow’s lab” is not spam.

Possible focus: misspellings, abbreviations, or legitimate messages containing words such as “free”. Investigate which method handles your proposed difficulty better.

Suggested data: UCI SMS Spam Collection

Option 2 Review Sentiment Analysis

Build a notebook that accepts a short review and returns Positive or Negative. For example, “The food was excellent” is positive, while “The food was not good” is negative. Use reviews with a clear positive or negative opinion and explain the limits of these two labels.

Possible focus: negative wording such as “not”, short reviews, or a selected review domain such as restaurants or products.

Suggested data: UCI Sentiment Labelled Sentences

Option 3 News Topic Classification

Build a notebook that accepts a headline or short news text and returns World, Sports, Business, or Science and Technology. For example, “The team won the football final” belongs to Sports.

Possible focus: very short headlines, unfamiliar wording, or confusion between two news categories.

Suggested data: AG News

Option 4: Student-Proposed NLP Project

Build a notebook that accepts [type of text] and returns [prediction, category, or result]. For example, “[example input]” should be classified as or produce “[expected output].”

Possible focus:
[Describe the specific NLP challenge the group will investigate, such as short texts, informal language, spelling errors, sarcasm, negation, or mixed languages.]

Suggested data:
Find a suitable publicly available labelled dataset related to your chosen task. Include the dataset name and link in your proposal.

Your proposal must also explain the project purpose, input and output, dataset, two methods to compare, experimental question, and responsibilities of each group member. The project idea and dataset require instructor approval.

General Project Requirements and Proposal

Proposed challenge profile and experimental question

Each group must propose its own challenge profile and experimental question within the selected project framework. The instructor will review and approve them to ensure feasibility and sufficient distinction from other groups.

Your challenge profile identifies the text or difficulty to study. Your question states what you will compare, for example: “Which method detects misspelled spam messages more reliably?”

Initial proposal and first draft

In Week 7, submit a one page proposal with group names and IDs. Cover your selected option, intended input and output, dataset and approximate size, two methods, proposed challenge, question and member responsibilities. Add a short paragraph discussing two relevant studies or technical references and cite them. Submit this first draft as Word or PDF on Blackboard; it carries 2 marks.

Proceed after instructor approval and revise the proposal if requested. Groups may use the same general model but must produce and explain their own investigation.

Implementation and evaluation

Data. Use an approved public labelled. State the source, labels, class counts and cleaning steps. English is the default; propose another language only with suitable data.

Two methods. Compare one traditional method with a method using a pretrained language model. A practical route is TF-IDF word features with logistic regression versus pretrained sentence representations with a classifier. For question pairs, word similarity versus pretrained sentence similarity is acceptable. Use libraries and pretrained weights as appropriate.

Fair testing. Use separate training, development and test examples. Learn from training data, choose settings on development data, and keep test data for the final comparison. Keep duplicates and linked question pairs in the same split. Compare both methods on identical test examples using accuracy and macro F1, which gives each class equal weight.

Analysis. Answer your approved question and explain at least three wrong predictions. If there are fewer errors, explain all of them and add difficult correct cases. An unsuccessful improvement can still support a strong analysis.

Individual work. Every member makes a technical contribution, supplies two original labelled examples, and keeps dated evidence of their work. Show original examples separately from test scores and complete the appendix. Credit all reused code, data, models and AI assistance; do not exchange completed work between groups.

Report Template and Project Report Outline

Report template and formatting

Use an IEEE manuscript template or another clear, consistent academic report format. Credit an external template if used. Include the title, course and section, group number, names and IDs, labelled tables or figures, and references. Page limit 8 pages.

Use the following seven sections. Describe the work you actually implemented and tested.

I Introduction

Introduce the NLP task, its practical use and intended users. State the selected project option and the purpose of your system.

II Project Goals and Objectives

State the proposed and approved challenge profile and experimental question. Describe the input, output and what the investigation is intended to demonstrate.

III Related Works

Briefly discuss at least two relevant studies or technical references. Explain how they informed your choices and how your investigation relates to them. Cite the original sources.

IV Data Acquisition and Preprocessing

Describe the dataset source, size, labels and class counts. Explain cleaning and text preparation, how examples were divided for training, development and testing, and how duplicate or related examples were kept together.

V Model Development and Training

Explain the traditional and pretrained model methods and justify their use. Describe training or model loading, any settings selected on development data, and the steps from text input to prediction. State which parts use existing libraries or weights.

VI Evaluation and Analysis

Present both methods' accuracy and macro F1 on the same test set. Use results to answer the question, explain three errors or the stated alternative, and discuss limitations. Include examples of the notebook accepting new input.

VII Conclusion and Future Work

Summarise the goal, main findings and practical usefulness. Suggest one realistic improvement based on the results. Keep claims proportionate to the evidence.

Add a reference list and attach the completed team appendix. The appendix records individual contributions and skills learned; it supports the individual assessment.

Submitting Your Source Code

Submit the complete implementation on **GitHub** so the instructor can inspect your methods, reproduce your results and test the system on new text. The notebook is the main code submission; include any supporting Python files it needs.

1 Make it runnable. State whether you use Google Colab or Jupyter. Include setup cells, package versions, data files or working download steps, and model-loading or training steps. Use relative file paths and tools within the approved computing resources.

2 Make it understandable. Use clear sections: Setup, Data, Method 1, Method 2, Results, Try it, and Sources. Explain key steps in short text cells and add useful code comments.

3 Make it usable. The Try it cell must accept a new message, review, news text or question pair and display a prediction from the selected model without retraining. Handle empty input with a helpful message.

4 Check before uploading. Restart the session and run every cell in order. Confirm that the results can be reproduced, the Try it cell works, and the report and slides show the same results. Keep output cells visible in the submitted notebook.

Submitting Your Work

Upload one ZIP file named GroupID_Project.zip to Blackboard containing:

1 The completed appendix with group details, contributions and skills learned.

2 The project report in Word or PDF format, using the seven-section outline.

3 The PowerPoint presentation, normally up to 5-10 slides covering the problem, data, methods, results and conclusion.

4 The working .ipynb notebook and all supporting code or files needed to run it. **Also, a GitHub Link**.

Give a 12 minutes group presentation including a live notebook demonstration. Every member must speak. Allow approximately two minutes of individual questions per student afterwards.

Assessment Due Dates

Stage	Course week and dates in 2026	Required work
Initial proposal and first draft	Week 7 4–8 October	Idea, question, intended output, brief related work, data and method plan.
Final submission	Week 14 29 November–3 December	Appendix, report, slides, notebook and supporting files.
Presentation and questions	Week 15 6–10 December	Group presentation, live demonstration and individual questions.

Exact submission times and presentation slots are on Blackboard. The course rules on late submission and academic integrity apply.

Assessment Deliverables and Grading

Group criteria total 20 marks. Presentation and questions (3) and contribution and teamwork (2) are marked individually. Each student receives the group score plus their individual score.

Assessment phase	Deliverable and marking checks	Marks	CLO
Initial proposal and first draft	Focused idea, question and intended output (1); brief related work and a feasible data and method plan (1).	2	1.1, 2.4
Presentation and questions Individual	Clear contribution to the presentation or demo (1); explains own code or method (1); explains a result or error during questions (1).	3	1.1, 2.4
Consistency of report format	Consistent headings, font and layout (1); clear table or figure labels and consistent citations (1).	2	2.4
Clarity and organisation of the report	Logical and understandable explanation (1); results and conclusions agree with the notebook evidence (1).	2	2.4
Quality of the investigation	Correct traditional method (1); correct pretrained model method (1); fair test with no training/test overlap (1); correct accuracy and macro F1 for both (1); answers the question and explains errors (1).	5	2.1, 2.2
Relevance and usefulness of the project	Clear practical task and target user (1); justified usefulness and realistic limitations (1).	2	1.1, 2.1
Quality of the source code	Model predicts from new input without retraining (1); readable output and empty-input handling (1); clear functions and useful comments (1).	3	2.3
Reusability and reproducibility of the source code	Setup and Run all work from a fresh session (2); clear data, model and package instructions (1); saved results can be reproduced (1).	4	2.3
Individual contribution and teamwork Individual	Verified technical contribution (1); evidence of cooperation and shared responsibility (1), supported by the appendix and dated work.	2	3.1

Total 25 marks

For a one-mark check, award 1 when demonstrated, 0.5 when partly demonstrated, and 0 when missing or incorrect. For the two-mark fresh-session check, award 2, 1 or 0 on the same basis. There is no required accuracy threshold; correct methods, fair testing and justified conclusions determine the mark.

CLO key: 1.1 explain NLP concepts and challenges; 2.1 apply NLP methods; 2.2 analyse traditional and modern methods; 2.3 develop NLP applications; 2.4 communicate orally and in writing; 3.1 work effectively in a team. These codes follow the updated course syllabus.

Appendix

Complete this appendix and submit it with the final report. Each member should identify their own technical work and explain the skills learned.

Course and section _____ Group number _____

Project option _____ Project title _____

Approved experimental question _____

_____

Team Leader

Name _____ Student ID _____

Technical contribution and dated evidence of work

_____

Skills learned and how I helped the team

_____

_____

Team Member 2

Name _____ Student ID _____

Technical contribution and dated evidence of work

_____

Skills learned and how I helped the team

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_____

Team Member 3

Name _____ Student ID _____

Technical contribution and dated evidence of work

_____

Skills learned and how I helped the team

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The team leader coordinates submission. All members share responsibility for the work and must be able to explain the system. Add space when completing this appendix electronically if needed.